BCS-093	Mobile Computing	L T P 3 0 0	3 Credits

Pre-requisites: Students should know about the basics of networking.

Course Objectives:

CO1	To learn about the basic concepts of Mobile Computing.
CO2	To understand about networking concepts relevant to modern wireless systems.
CO3	To introduce emerging mobile computing ideas and best practices
CO4	To gain hands-on knowledge practice with mobile computing
CO5	To understand various types of Protocols

Detailed Syllabus

MODULE-I

Introduction: Issues in mobile computing, overview of wireless telephony: cellular concept, GSM: air-interface, channel structure, location management: HLR-VLR, handoffs, channel allocation in cellular systems, CDMA, GPRS. Wireless LAN Overview: MAC issues, IEEE 802.11, Blue Tooth, Wireless multiple access protocols, TCP over wireless, Wireless applications, Mobile IP, WAP: Architecture, protocol stack, application environment,

MODULE-II

Data management issues: data replication for mobile computers, adaptive clustering for mobile wireless networks, File system, Disconnected operations. Mobile Agents computing, security and fault tolerance, transaction processing in mobile computing environment.

MODULE-III

Adhoc networks and localization: Adhoc Networks issues, Routing protocols, global state routing (GSR), Destination sequenced distance vector routing (DSDV), Dynamic source routing (DSR), AdHoc on demand distance vector routing (AODV), Temporary ordered routing algorithm (TORA), QoS in Ad Hoc Networks, applications.

Course Outcomes: After the completion of the course the student will be able to:

CO1	Various wireless communication technologies.
CO2	Enables the students to visualize the various important steps in GSM communication
CO3	To acquire the knowledge of the fuzzy Neural network and Genetic Language
CO4	Enables the students to analyze the mobile IP and Transport Protocol.
CO5	To analyze various type of Protocols

Text Books:

1. J. Schiller, Mobile Communications, Addison Wesley.
2. Charles Perkins, Mobile IP, Addison Wesley.

Reference Books:

1. Charles Perkins, Ad hoc Networks, Addison Wesley.
2. Upadhyaya, "Mobile Computing", Springer